

Lattice Cryptography

An Elementary Introduction

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1. Vectors

A *vector* is an arrow. It tells you how far to move sideways and how far to move up or down.

Throughout this paper we use two vectors, $\mathbf{r}_1 = (3, 1)$ and $\mathbf{r}_2 = (-1, 2)$. The first number is the sideways step and the second is the vertical step. So $\mathbf{r}_1$ means 3 steps right and 1 step up, and $\mathbf{r}_2$ means 1 step left and 2 steps up.

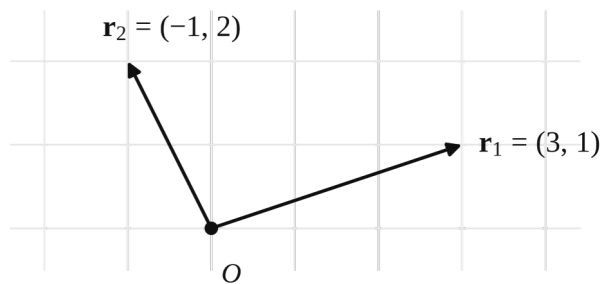


Figure 1. The vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ drawn from a starting point O .

2. Lattices

Starting from O , we may use each vector as many times as we like, forward or backward. Only whole steps are allowed; there are no half steps. For example, taking $\mathbf{r}_1$ three times and then $\mathbf{r}_2$ once gives $3\mathbf{r}_1 + \mathbf{r}_2 = (8, 5)$.

A *lattice* is the set of every point that can be reached this way. It is a pattern of dots that repeats forever in every direction. It makes a grid of dots in the plane.

A pair of vectors that builds a lattice is called a *basis* for that lattice. You can think of a basis as an ingredient for a lattice, and vectors are ingredients for bases.

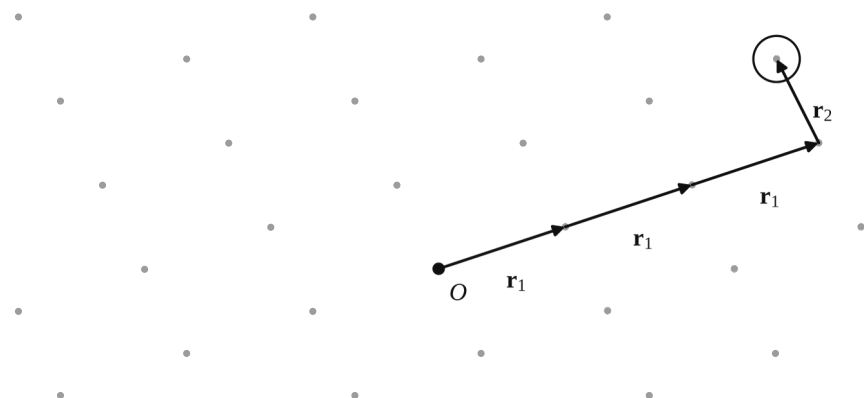


Figure 2. The path $3\mathbf{r}_1 + \mathbf{r}_2$ reaches the circled point. Every dot shown can be reached with some combination of whole steps.

3. The Closest Vector Problem

Suppose we are given a point C that does not lie on the lattice (it's not one of the dots on the grid).

We want the lattice point nearest to it, together with the steps that reach it.

Given a basis and a point C , find the lattice point closest to C .

With the basis $\mathbf{r}_1, \mathbf{r}_2$ the answer is easy to find: move backward along $\mathbf{r}_1$ twice and forward along $\mathbf{r}_2$ twice. That is, $-2\mathbf{r}_1 + 2\mathbf{r}_2 = (-8, 2)$.

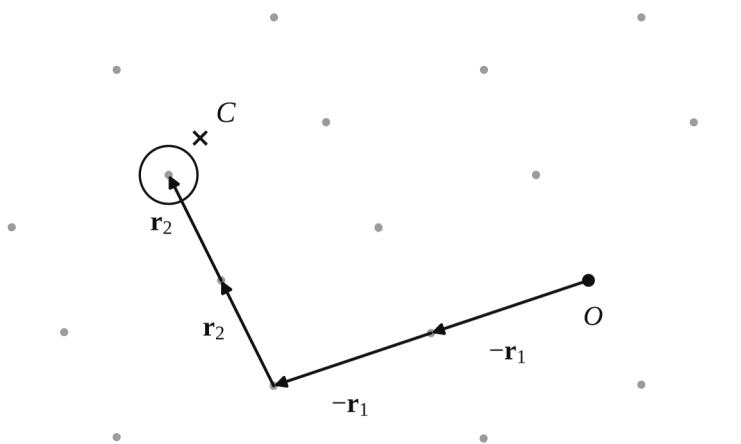


Figure 3. The path $-2\mathbf{r}_1 + 2\mathbf{r}_2$ reaches the circled point, the lattice point closest to C .

4. Different Bases for the Same Lattice

A lattice has more than one basis. Consider $\mathbf{b}_1 = (5, 4)$ and $\mathbf{b}_2 = (8, 5)$. Using whole steps of these two vectors, we reach exactly the same dots as before, no more and no fewer. This holds because

$\mathbf{b}_1 = 2\mathbf{r}_1 + \mathbf{r}_2$ and $\mathbf{b}_2 = 3\mathbf{r}_1 + \mathbf{r}_2$, and the steps can be undone: $\mathbf{r}_1 = -\mathbf{b}_1 + \mathbf{b}_2$ and $\mathbf{r}_2 = 3\mathbf{b}_1 - 2\mathbf{b}_2$. The vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ are long and point in almost the same direction. We call $\mathbf{r}_1, \mathbf{r}_2$ a *good basis* and $\mathbf{b}_1, \mathbf{b}_2$ a *bad basis*.

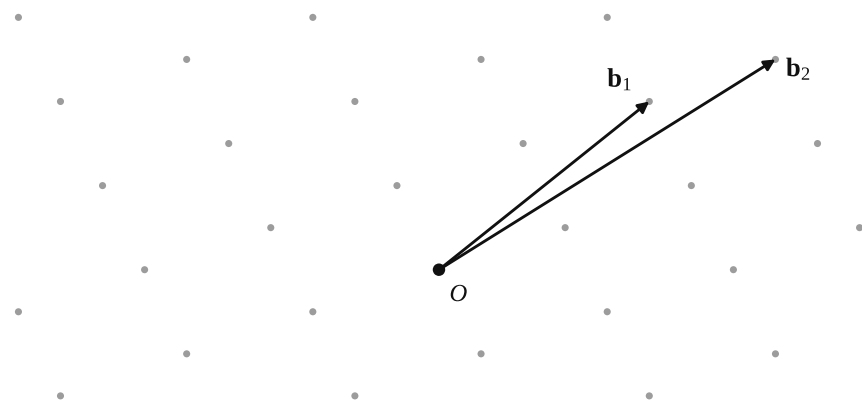


Figure 4. The bad basis $\mathbf{b}_1, \mathbf{b}_2$ builds the same lattice as $\mathbf{r}_1, \mathbf{r}_2$.

5. Why a Bad Basis Makes the Problem Hard

Now solve the same problem with $\mathbf{b}_1$ and $\mathbf{b}_2$. A step that moves closer to C in one direction moves away from it in another, so every path zigzags. The answer is $8\mathbf{b}_1 - 6\mathbf{b}_2 = (-8, 2)$: fourteen long steps to reach the same point that took four short steps before.

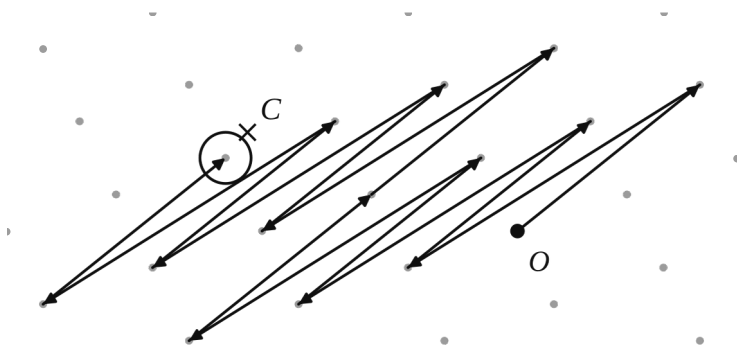


Figure 5. The path $8\mathbf{b}_1 - 6\mathbf{b}_2$. The destination is the same as in Figure 3, but it is much harder to find.

Remark 1. The lattice is the **EXACT SAME** in both cases; only the basis has changed. A good basis makes the Closest Vector Problem easy. A bad basis makes it hard.

6. Higher Dimensions

A page has two directions: left–right and up–down. The lattices used in real encryption have about 1,000 directions, called *dimensions*. They cannot be drawn, but the same rules apply. Two facts make the problem hard in high dimensions.

- (i) We are given only the basis, never the full set of dots. After finding a lattice point near C , we must still check every nearby point to confirm it is the closest.
- (ii) The number of nearby points grows very quickly with dimension: roughly like $r \times r$ in two dimensions, $r \times r \times r$ in three, and so on. In 1,000 dimensions, a step that helps in one direction can hurt in the other 999.

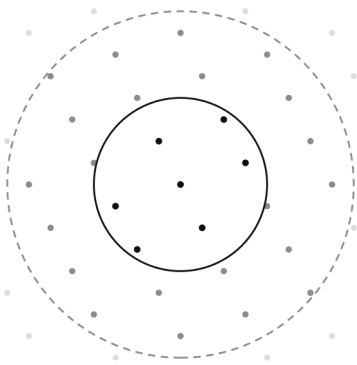


Table 1. Factor by which the number of points grows when the search radius is doubled.

Dimensions	Points to check grow by
1	2
2	4
3	8
10	1,024
100	2^{100} , a 31-digit number
1,000	2^{1000} , a 302-digit number

Figure 6. The solid circle (radius 4) contains 7 lattice points; the dashed circle (radius 8) contains 29. Doubling the distance gives about 4 times as many points.

7. Public and Private Keys

Each person who wants to receive messages chooses a lattice and two bases for it:

- (i) a *private key*: a good basis, kept secret;
- (ii) a *public key*: a bad basis, shared with everyone.

Both keys describe the same lattice, but only the private key makes the Closest Vector Problem easy.

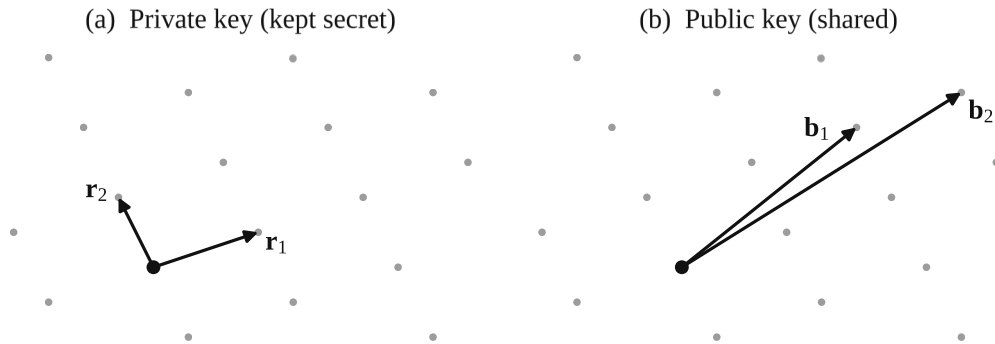


Figure 7. One lattice described by two different bases.

8. Encrypting a Message

Suppose we wish to send the number “7” to a recipient. We proceed in three steps.

1. Using the recipient's public key, choose a lattice point that stands for 7.
2. Add a small random displacement, called *noise*. The result is near the chosen point but not on it.
3. Send the result. Anyone may see it.

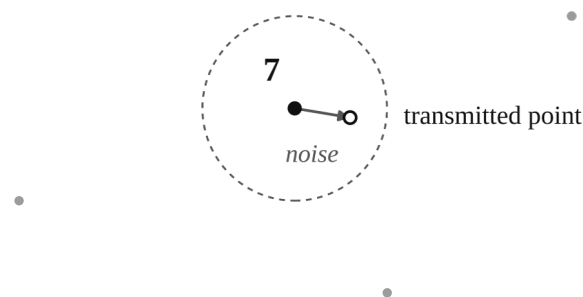


Figure 8. The lattice point standing for 7, the noise, and the transmitted point. The noise is small, so the transmitted point stays inside the dashed circle and is still closest to the point for 7.

9. Decrypting a Message

To decrypt, the recipient solves the Closest Vector Problem: find the lattice point nearest the received point. That point stands for 7.

With the private key there is a shortcut. Write the received point as an amount of $\mathbf{r}_1$ plus an amount of $\mathbf{r}_2$. The amounts come out as decimals, 1.16 and 0.87. Rounding to whole numbers gives 1 and 1, and $\mathbf{r}_1 + \mathbf{r}_2$ is the correct point.

An eavesdropper has only the public key. The same method with $\mathbf{b}_1$ and $\mathbf{b}_2$ gives 1.46 and -0.59 . Rounding gives 1 and -1 , and $\mathbf{b}_1 - \mathbf{b}_2 = (-3, -1)$, which is the wrong point.

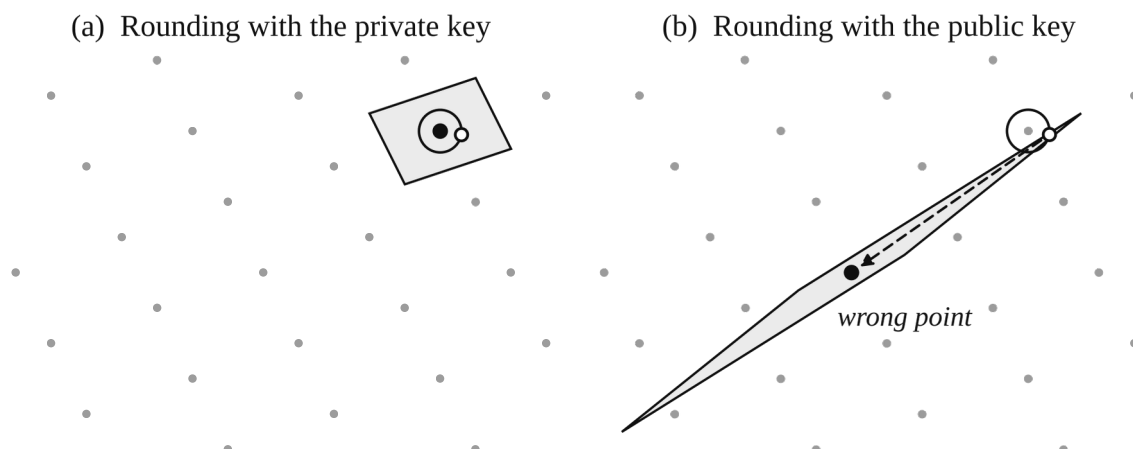


Figure 9. Rounding sends the received point (open circle) to the center of the shaded tile that contains it. (a) The private key gives small, nearly square tiles, so rounding reaches the correct point (ringed). (b) The public key gives long, thin tiles, so rounding lands far away.

In 2 dimensions, a computer could still check every nearby point and find the error. In 1,000 dimensions, no fast method is known without the private key.

10. Security Against Quantum Computers

Much of today's internet is protected by methods such as RSA. These rely on problems like splitting a large number into its factors. Ordinary computers cannot solve such problems in any reasonable time, but a large enough quantum computer could.

As far as is currently known, the Closest Vector Problem in about 1,000 dimensions is extremely hard for both ordinary and quantum computers. For this reason lattices are a leading choice for *post-quantum* encryption. In 2024 the U.S. National Institute of Standards and Technology (NIST) adopted a lattice-based method, ML-KEM, as a federal standard.

Remark 2. The scheme described here is simplified. Practical systems use a closely related problem called *Learning With Errors*, but the central idea is the same: a lattice, a small amount of noise, and a secret good basis.

11. Significance of the Proof

Although these concepts may seem complex, they ultimately reduce to **fundamental arithmetic operations**: addition, subtraction, multiplication, and division.

Just as an exponent like 3^2 expands to

$$3 \times 3$$

and ultimately to

$$3 + 3 + 3,$$

we can apply a similar expansion to the underlying mathematics of **lattice cryptography**.

However, **digital systems** require more than basic arithmetic. Because lattice cryptography runs on digital hardware (computers), we must express these mathematical operations in a format native to **computer architecture**.

This requires deconstructing operations such as addition down to their physical implementation as **electronic states** within a digital system down to the **atomic level**.

Because direct physical analysis of these atomic interactions requires instruments costing millions of dollars, such as atomic-resolution transmission electron microscopes [1, 2], we instead model this physical realm using precise **mathematical notation** and **formal proofs**.

In the introduction, a vector was represented by a simple ordered pair: (x, y)

But to a computer, (x, y) means nothing. We instead have to use rigorous definitions such as:

$$\vec{x} = \left(\text{rem}(\Xi[(P_{a_1}, P_{b_1}), (P_{c_1}, P_{d_1})]), \text{rem}(\Xi[(P_{a_2}, P_{b_2}), (P_{c_2}, P_{d_2})]), \dots, \text{rem}(\Xi[(P_{a_u}, P_{b_u}), (P_{c_u}, P_{d_u})]) \right).$$

More complicated concepts must use definitions such as:

$$\overrightarrow{A \vec{x}} = \left(\Gamma \left(\Lambda^{-1}(\Upsilon[(P_{a_1}, P_{b_1}), \Lambda^{-1}(x_1)]), \Lambda^{-1}(\Upsilon[(P_{c_1}, P_{d_1}), \Lambda^{-1}(x_2)]) \right), \Gamma \left(\Lambda^{-1}(\Upsilon[(P_{a_2}, P_{b_2}), \Lambda^{-1}(x_1)]), \Lambda^{-1}(\Upsilon[(P_{c_2}, P_{d_2}), \Lambda^{-1}(x_2)]) \right) \right).$$

Why do this?

By reducing complex cryptographic concepts to their foundational physical and computational representations, we can derive mathematical concepts, formulas, and functions defined and discussed in this proof that do not only represent math “on paper”, but a clear map of how computers execute lattice cryptography at the **atomic hardware level**.

While lattice cryptography operates within digital systems, reducing these concepts to their foundational representations bridges the gap between **hardware execution** and **mathematical abstraction**. Ultimately, the primary goal of this research is to uncover and understand the **physical cryptographic secrets**, concepts, and phenomena that underlie and govern the digital system.

References

- [1] Lawrence Berkeley National Laboratory, “Twenty Years and Still Growing (Smaller),” *Science Beat*, Jan. 9, 2004. The lab’s 1.6 Å Atomic Resolution Microscope was contracted for \$3 million. <https://www2.lbl.gov/Science-Articles/Archive/sb-NCEM-20.html>
- [2] University of Illinois Chicago, “NSF awards \$4M to UIC to home world’s first analytical aberration-corrected and monochromated transmission electron microscope with magnetic field-free objective lens,” Sept. 28, 2022. <https://las.uic.edu/news-stories/nsf-awards-4m-to-uic-to-home-worlds-first-analytical-aberration-corrected-and-monochromated-transmission-electron-microscope-with-magnetic-field-free-objective-lens>